Easy (1–10)
Read and print a 2D array of size m × n.
Find the sum of all elements in a 2D array.
Find the largest element in a 2D array.
Find the smallest element in a 2D array.
Calculate the sum of each row.
Calculate the sum of each column.
Count the number of even and odd elements in a matrix.
Search for a given element in a matrix and print its position.
Print only the main diagonal elements of a square matrix.
Print only the secondary diagonal elements of a square matrix.

Medium (11–20)
Find the transpose of a matrix.
Check whether a given matrix is symmetric.
Add two matrices.
Subtract two matrices.
Multiply two matrices.
Find the sum of the main diagonal and secondary diagonal separately.
Check whether a matrix is an identity matrix.
Check whether a matrix is a diagonal matrix.
Print the boundary elements of a matrix.
Print the matrix in spiral order.

Hard (21–30)
Rotate a square matrix by 90° clockwise without using another matrix.
Rotate a square matrix by 90° anticlockwise without using another matrix.
Find the saddle point of a matrix (smallest in its row and largest in its column).
Print all elements in zigzag (wave) order.
Find the row having the maximum sum.
Find the column having the minimum sum.
Check whether a matrix is a sparse matrix.
Find the determinant of a 3 × 3 matrix.
Find the inverse of a 2 × 2 matrix.
Search an element in a matrix where every row and every column is sorted in ascending order with O(m + n) time.

Bonus Challenge Questions (Interview Level)
Print matrix in snake pattern.
Print upper triangular matrix.
Print lower triangular matrix.
Check whether two matrices are equal.
Check whether a matrix is upper triangular.
Check whether a matrix is lower triangular.
Find the frequency of every element in a matrix.
Replace all negative elements with zero.
Sort each row of the matrix.
Sort each column of the matrix.
Find the second largest element in the matrix.
Find common elements between two matrices.
Print all neighbors of a given cell.
Find the maximum difference between any two elements where the larger element comes after the smaller one.
Implement the Game of Life (advanced).